

Bridging the gap between collisional and collisionless plasma shocks: A simulation study using OSIRIS

Cite as: Phys. Plasmas **33**, 022112 (2026); doi: [10.1063/5.0294460](https://doi.org/10.1063/5.0294460)

Submitted: 4 August 2025 · Accepted: 2 February 2026 ·

Published Online: 24 February 2026



View Online



Export Citation



CrossMark

Yossef Nissim Kindi,^{1,a)} Asaf Pe'er,¹ Antoine Bret,^{2,3} Luís O. Silva,⁴ and Kevin M. Schoeffler^{4,5}

AFFILIATIONS

¹Bar Ilan University, Ramat Gan 5290002, Israel

²ETSI Industriales, Universidad de Castilla-La Mancha, 13071 Ciudad Real, Spain

³Instituto de Investigaciones Energéticas y Aplicaciones Industriales, Campus Universitario de Ciudad Real, 13071 Ciudad Real, Spain

⁴GoLP/Instituto de Plasmas e Fusão Nuclear, Instituto Superior Técnico, Universidade de Lisboa, 1049-001 Lisbon, Portugal

⁵Institut für Theoretische Physik, Ruhr-Universität Bochum, D-44780 Bochum, Germany

^{a)}Author to whom correspondence should be addressed: ykindi@gmail.com

ABSTRACT

Shock waves in plasma can be characterized according to the physical mechanisms behind their formation. When collisions are frequent, shock dissipation is mainly binary-collision driven, and the shock width (SW) is often comparable to a few mean-free-paths. In contrast, collisionless shocks rely on collective plasma processes to establish dissipation on scales far below the mean-free-path. The purpose of this study is to bridge these two regimes. Here, we present simulations using the OSIRIS particle-in-cell code with the Coulomb collision module to explore the gradual transition between collisional and collisionless shocks by varying plasma parameters that determine collisionality. We show a smooth transition of the SW in the intermediate region, where the ion plasma parameter is $N_D \approx 1$. Our numerical results confirm earlier theoretical predictions from existing theories in the asymptotic regimes, where the Mott-Smith ansatz, combined with a full Bhatnagar-Gross-Krook (BGK) collision operator, was used to obtain the collisional-regime SW, while the classical Tidman formalism describes the collisionless limit. We demonstrate that the ion plasma parameter provides a useful metric for identifying when shocks transition from a fluid-like, mean-free-path scale to a collisionless, sub-mean-free-path scale. We discuss the importance of our results in astrophysical environments, where shock breakout causes a transition in the SW and marks the beginning of the acceleration of particles to high energies.

© 2026 Author(s). All article content, except where otherwise noted, is licensed under a Creative Commons Attribution (CC BY) license (<https://creativecommons.org/licenses/by/4.0/>). <https://doi.org/10.1063/5.0294460>

I. INTRODUCTION

Shock waves are commonly found in nature in systems that involve fluid flows. Such systems exist on very different scales, ranging from the microphysical scale to the astronomical scale. As a result, the properties of the shocks can vary considerably. Conceptually, shock waves in plasma can be divided into two main categories: *collisional* and *collisionless*. In a collisional fluid, momentum and energy dissipation occur through frequent binary collisions of the constituent particles, causing shocks to form over scales of a few mean-free-paths. This phenomenon has been extensively studied in neutral fluids, and it remains valid as long as collisions are sufficiently frequent to thermalize particle velocities locally and drive the plasma to a near-Maxwellian distribution.^{1,2}

In contrast, *collisionless* shocks are sustained by collective interactions such as electrostatic or electromagnetic (EM) instabilities, wave-particle interactions, and self-regulated electric and magnetic fields.^{3–5} By definition, the collisional mean-free-path in a collisionless regime is very large compared to the system length scale; yet shocks can still form when upstream bulk kinetic energy is dissipated by collective plasma effects. Classic examples include the Earth's bow shock in the solar wind—measured to be orders of magnitude thinner than the proton mean-free-path,^{5,6} as well as shocks in supernova remnants and gamma-ray bursts.^{7,8}

Bret and Pe'er⁹ attempted to unify the understanding of collisional and collisionless shocks. They did that by using the Mott-Smith ansatz¹⁰ and the full Bhatnagar-Gross-Krook (BGK) collision term¹¹

to derive a shock width (SW) in the collisional regime, and Tidman's approach¹² in the collisionless one. This approach describes a shock mediated by electrostatic turbulence on scales much smaller than the mean-free-path. In that work, Bret and Pe'er⁹ identified the relevant regime using a single parameter that controls the transition, namely, the ion plasma parameter, N_D ,

$$N_D = \frac{4\pi}{3} \lambda_{Di}^3 n_0. \quad (1)$$

Here, n_0 is the particle density and λ_{Di} is the ion Debye length defined as $\sqrt{\frac{\epsilon_0 k_B T_i}{n_0 q_e^2}}$, where ϵ_0 is the vacuum permittivity, k_B is the Boltzmann constant, T_i is the ion temperature, and q_e is the electron charge.¹³ Physically, N_D corresponds to the approximate number of particles inside one Debye sphere, and it is directly related to the coupling strength of the plasma. For large $N_D > 1$, the collision rate becomes negligible compared to the plasma frequency, and kinetic energy dissipation must be mediated by collective plasma effects. On the other hand, for small N_D , binary collisions dominate, giving rise to fluid-like collisional behavior.

The approach used by Bret and Pe'er⁹ was analytic, using the above-mentioned assumptions, which are valid from the asymptotic limits of $N_D \gg 1$ and $N_D \ll 1$ up to the transition between the two regimes. It is therefore of interest to examine these assumptions, as well as provide proper calculations in the regime of $N_D \approx 1$. Related kinetic studies of collisional plasmas have also employed single-parameter measures of collisionality to connect collisionless and collisional behavior. For example, Zheng *et al.*¹⁴ analyzed how ion-ion collisions modify the dispersion and damping of ion-acoustic-type modes, and hence the Thomson-scattering spectra, in homogeneous plasmas. In this paper, we present a set of simulations that aim to bridge the gap between these extremes. We employed the OSIRIS particle-in-cell (PIC) code¹⁵ to model two interpenetrating plasma streams with varying densities and temperatures, which in turn modify the effective collisionality. By tracking the shock formation and measuring the width of the density jump, we investigate the continuous transition from a collisional to a collisionless shock. We compare our simulation results with theoretical expectations, demonstrating how the plasma parameter N_D offers a quantitative criterion to identify the shock's nature.

Taken together, these simulations provide a unified physical framework that continuously connects collisional and collisionless shocks using N_D as a control parameter, and the first systematic PIC-based verification, using a first-principles approach, of the shock-width scaling proposed by Bret and Pe'er across several orders of magnitude in N_D . In this way, our results complement previous collisional plasma studies based on mean-free-path-based collisionality measures, while focusing specifically on the self-consistent structure and thickness of shock fronts.

II. SIMULATION SETUP AND THE OSIRIS CODE

A. OSIRIS model

Our numerical study uses the OSIRIS PIC code,^{15,16} a framework that evolves Maxwell's equations on a spatial grid and tracks the trajectories of discrete charged particles. These particles are subject to the Lorentz force, and, at each time step, they are pushed to a new position and momentum according to the local electromagnetic (EM) fields

inside the relevant cell. This movement of particles will, in turn, affect the physical quantities in the cell, such as density, temperature, current, electric, and magnetic fields. OSIRIS updates at each time step both the physical quantities and the particle's position and momentum in all the grid cells.

Since collisions are a crucial element in our study, we use the OSIRIS version that includes a collision module.^{16,17} This module simulates collisions by randomly pairing particles within the same cell and assigning a scattering angle at each time step. Each pair undergoes an elastic, energy, and momentum conserving scattering in which the relative velocity $\mathbf{u} = \mathbf{v}_\alpha - \mathbf{v}_\beta$ is rotated by a small random angle θ . The rotation angle θ is determined by defining a deflection parameter,

$$\delta = \tan\left(\frac{\theta}{2}\right).$$

When $\delta \rightarrow 0$, the scattering angle approaches zero (no collisions), while $\delta \rightarrow \infty$ corresponds to a head-on encounter ($\theta \approx 180^\circ$).

The deflection parameter δ is randomly chosen from a zero-mean Gaussian distribution, and its variance is proportional to the physical Coulomb collision frequency. For a pair of charges q_α, q_β and reduced mass $m_{\alpha\beta} = m_\alpha m_\beta / (m_\alpha + m_\beta)$, the variance per time step Δt is [Ref. 18, Eq. (8a)]

$$\langle \delta^2 \rangle = \frac{q_\alpha^2 q_\beta^2 n_L \ln \Lambda}{8\pi \epsilon_0^2 m_{\alpha\beta}^2 u^3} \Delta t. \quad (2)$$

Here, $n_L = \min(n_\alpha, n_\beta)$ is the lower of the two species densities, $\ln \Lambda$ is the Coulomb logarithm, and u is the magnitude of $\mathbf{u}$. Equation (2) ensures that the mean square deflection accumulated in time Δt is $\langle \theta^2 \rangle \propto \nu_{\alpha\beta} \Delta t$, when $\nu_{\alpha\beta}$ is the collision rate between the two species, reproducing the Landau collision operator in the small-angle limit.¹⁸ Thus, although collisions are applied at every PIC step, their strength automatically scales with the physical collision frequency $\nu_{\alpha\beta}$.

From Eq. (2), one sees that collisions become more frequent at a higher density and slower relative speeds between the particles, and the collision frequency scales inversely with the reduced mass. If more than two particles share a cell, they are grouped into pairs (or triplets, if an odd number remains) and scatter successively in one time step.

Originally proposed by Takizuka and Abe,¹⁸ this procedure has been refined for better computational efficiency by aggregating multiple small-angle collisions (within a single cell) into a single, larger-angle deflection,¹⁹ and, in addition, has been extended to relativistic regimes.^{20,21} The OSIRIS code implements these adaptations,¹⁷ thereby enabling self-consistent modeling of both electromagnetic fields and Coulomb collisions in large-scale PIC simulations.

B. Simulation setup

We perform simulations in a two-dimensional (2D) grid, with a varying number of cells, ranging from 3328 to 16640 in the x-direction and 16 cells in the y-direction (quasi-one-dimensional simulation). The cells are resolved to the electron Debye length, i.e., $\Delta x \approx \Delta y \approx 0.3 \lambda_{De}$, where λ_{De} is the Debye length of the electrons, defined as $\sqrt{\frac{\epsilon_0 k_B T_e}{n_0 q_e^2}}$, where T_e is the temperature of the electrons. The initial electric and magnetic fields are zero in all simulations; while the initial temperature, density, and fluid velocities vary throughout the simulations, all are kept non-relativistic. All our simulations

consider electron-ion plasma, with a realistic mass ratio of 1836. To ensure good accuracy, we use current smoothing and 36 particles per cell for both electrons and ions, namely, $2 - 10 \times 10^6$ particles in total. The time steps of the simulation are measured in $1/\omega_p$, where ω_p is the plasma frequency defined as $\sqrt{\frac{n_0 q_e^2}{m_e \epsilon_0}}$, where m_e is the electron mass. All simulations ran for $4000 1/\omega_p$, which we find to be a sufficient time for the formation of a shock.

To form shocks, we initialize two semi-infinite slabs.²² One slab is injected with a drift speed v_0 toward a stationary slab. To ensure the development of a shock, the drift speed is such as to ensure a Mach number larger than 1. Table I lists the initial conditions of the simulation setups used. These include: density of particles (number per cm^3 , n_0); ratio between electron and ion temperature (R_T); electron temperature in Kelvin (T_e); relative drift velocity between the two slabs (v_0) in units of the speed of light (c); the ion plasma parameter, N_D ; the injection Mach number M_{inj} ; the shock Mach number M_{sh} ; and the resulting shock width (SW) in units of the electron inertial length, c/ω_p . The injection Mach number is defined as

$$M_{\text{inj}} \equiv \frac{v_0}{v_s},$$

where v_s is the upstream sound speed of the incoming slab,²³

$$v_s = \sqrt{\frac{k_B T_e + k_B \gamma T_i}{m_i}}. \quad (3)$$

Here, m_i and γ are the ion mass and the adiabatic index, respectively. The shock Mach number M_{sh} is obtained *a posteriori* by tracking the position of the density jump as a function of time, transforming to the downstream frame, and dividing the shock speed by the local upstream sound speed immediately ahead of the front. In this way, the estimate of M_{sh} automatically includes the modest increase in the upstream temperature produced during the evolution. For the representative

runs where we measured it, $M_{\text{sh}} \simeq 1.6$ – 1.7 , close to $v_0/2v_s$, as expected for symmetric counterstreaming slabs (see Table I). We further included to Table I the resulting shock width (SW) in units of the electron inertial length, c/ω_p .

1. Run procedure

We allow each simulation to evolve until two well-defined shock fronts form near the interface between the two slabs. We then track the density profile along the transition from upstream to downstream plasma. The shock width ℓ is measured by locating where the density rises from $\sim 10\%$ to $\sim 90\%$ of the downstream density. This approach is repeated for the entire parameter space to produce a comprehensive view of how collisionality affects shock structure.

III. RESULTS: SHOCK WIDTHS IN COLLISIONAL AND COLLISIONLESS PLASMAS

A. Shock wave formation

We identify the formation of shock waves via the abrupt change in the ion density, resulting from the interpenetration of the plasma slabs at a supersonic speed. Figure 1 shows the temporal evolution of the ion density (top panels) and of the corresponding ion phase-space distribution (bottom panels) for a representative run in the intermediate collisionality regime with $N_D = 2.78$, at $t = 1120$, 2400 , and $3520 \omega_p^{-1}$. The two shock fronts, located near $x \simeq 40$ – $60 c/\omega_p$, are clearly visible, together with a wavetrain in the downstream region that develops and extends as the system evolves. The observed oscillations are likely electrostatic ion-acoustic waves, which can be identified and characterized by comparing the density ripples with the modulations in the ion phase-space density. This comparison shows that the ion-acoustic wavetrain remains essentially linear, while the ions behind the shock do not relax to a fully Maxwellian state but retain a non-thermal structure in phase space.

TABLE I. Parameters of the OSIRIS simulation runs performed. The parameters were chosen to represent different collisionalities. From left to right are the particle density n_0 (cm^{-3}); R_T , the temperature ratio between electrons and ions (T_e/T_i); electron temperature T_e (K); drift velocity v_0 in units of the speed of light c ; ion plasma parameter N_D ; injection Mach number $M_{\text{inj}} = v_0/v_s$; shock Mach number M_{sh} measured from the simulations; and the resulting shock width (SW) in units of the electron inertial length c/ω_p .

n_0	R_T	T_e	v_0	N_D	M_{inj}	M_{sh}	SW
1×10^{22}	400	2.97×10^7	0.0066	2.78×10^{-1}	3.99	1.73	0.50
1×10^{22}	9	2.97×10^7	0.0066	8.25×10^1	3.67	1.3	0.73
1×10^{16}	9	2.97×10^7	0.0066	8.25×10^4	3.67	1.25	0.77
1×10^{18}	9	2.97×10^7	0.0066	8.25×10^3	3.67	1.28	0.88
1×10^{20}	9	2.97×10^7	0.0066	8.25×10^2	3.67	1.23	0.66
1×10^{22}	400	1.07×10^7	0.0040	6.01×10^{-2}	4.03	1.71	0.38
1×10^{20}	400	1.07×10^7	0.0040	6.01×10^{-1}	4.03	1.77	0.29
1×10^{18}	400	1.07×10^7	0.0040	6.01×10^0	4.03	1.75	0.29
1×10^{19}	400	1.07×10^7	0.0040	1.90×10^0	4.03	1.76	0.24
1×10^{21}	400	1.07×10^7	0.0040	1.90×10^{-1}	4.03	1.87	0.28
1×10^{20}	400	2.97×10^7	0.0066	2.78×10^0	3.99	1.63	0.53
1×10^{18}	400	2.97×10^5	0.0007	2.78×10^{-2}	4.23	1.44	0.06
1×10^{19}	400	2.97×10^5	0.0007	8.81×10^{-3}	4.23	1.1	0.07
1×10^{16}	400	2.97×10^7	0.0066	2.78×10^2	3.99	1.70	0.58
1×10^{22}	400	1.07×10^7	0.0066	6.01×10^{-2}	6.66	2.13	0.26

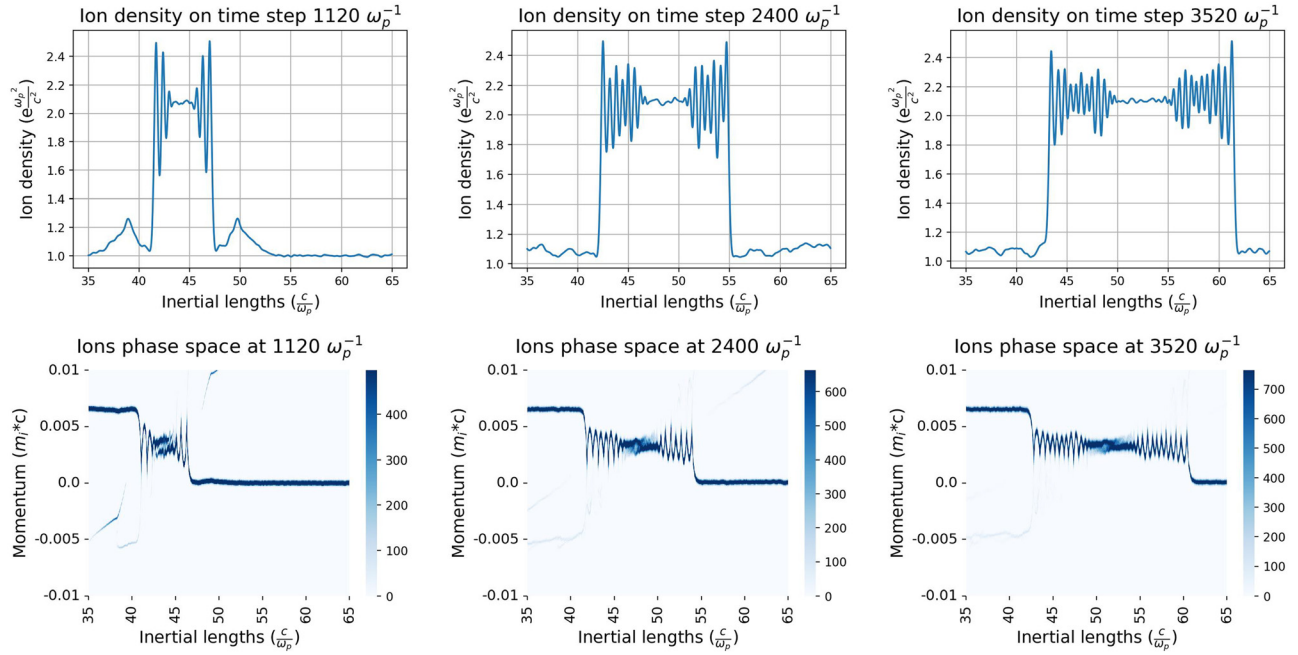


FIG. 1. Temporal evolution of the shock structure for a run with ion plasma parameter $N_D = 2.78$. Top row: ion density profiles at $t = 1120, 2400$, and $3520 \omega_p^{-1}$, showing the formation of two shocks and the growth of a downstream wavetrain. Bottom row: corresponding ion phase-space distributions, which reveal the ion-acoustic wave activity and the gradual relaxation of the ions behind the shock.

To further illustrate how the degree of collisionality affects the ion distribution, we present in Fig. 2 a direct comparison of the late-time ion phase-space structure (p_1-x_1) for three representative regimes: a collisional shock ($N_D = 0.06$), an intermediate case ($N_D = 2.78$), and a collisionless shock ($N_D = 278$). In the intermediate and collisionless regimes, the phase-space plots exhibit pronounced oscillatory structure and partial ion reflection, consistent with ion-

acoustic wavetrains and an electrostatic, collectively mediated shock. In contrast, the collisional case displays a smooth, thermally broadened distribution with no oscillatory signatures, demonstrating that strong binary scattering suppresses wave activity and results in a more fluid-like transition. Following the time evolution of these diagnostics for representative runs in each regime, we do not find any evidence for an initially collisional front that later turns into a purely electrostatic,

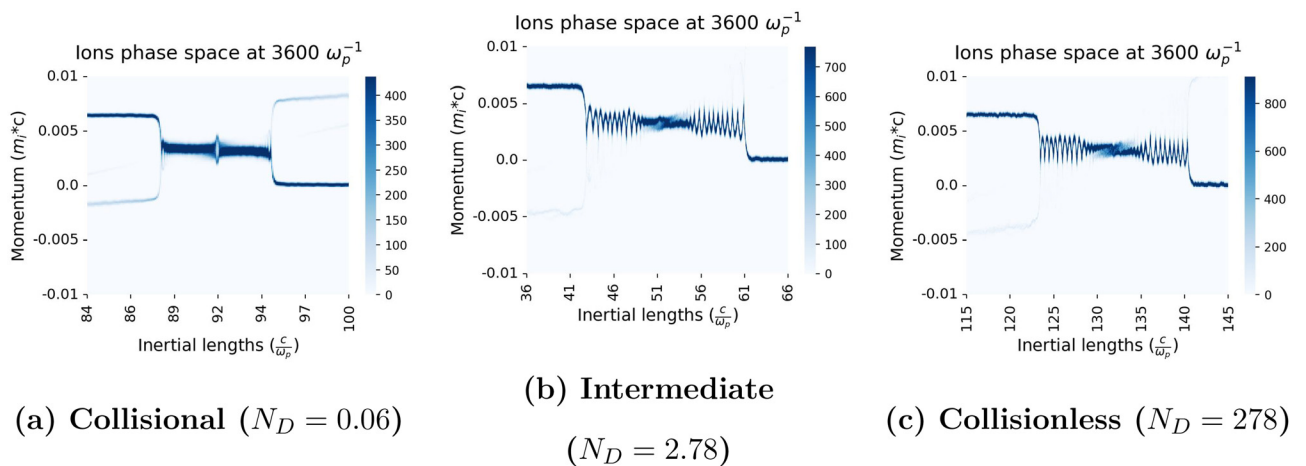


FIG. 2. Comparison of ion phase-space distributions (p_1-x_1) for representative collisional, intermediate, and collisionless shocks at late times. (a) The collisional case ($N_D = 0.06$) shows smooth thermal broadening with no oscillatory structure. (b) and (c) The intermediate ($N_D = 2.78$) and collisionless ($N_D = 278$) regimes both exhibit clear oscillations and hints of ion trapping, characteristic of ion-acoustic wave activity and collective electrostatic shock mediation.

wave-mediated structure; instead, the shock retains its collisional or collisionless character throughout the simulation.

B. Plasma parameter as an indication of the shock collisionality

We quantify plasma collisionality using the ion plasma parameter N_D . A large N_D signifies that collective plasma interactions overshadow binary collisions, leading to a more *collisionless* behavior, whereas small N_D indicates a stronger, *collisional* coupling. We estimate the ion collision rate as $\nu_{\text{coll}} = \nu_{\text{th}}/l_{\text{mfp}}$, where ν_{th} is a characteristic thermal velocity defined as $\sqrt{k_B T_i/m_i}$, and l_{mfp} is the mean-free-path. In the analytical estimates carried in this section, we consider ion–ion collisions only, so ν_{coll} and l_{mfp} refer to ion–ion scattering. We do stress that in the numerical results shown below, we consider interactions between all particles species: ion–ion, ion–electron, and electron–electron, although the contribution of the electrons is small for the temperature ratios chosen here, which does not exceed 400.²⁴ Our mean-free-path extended to $N_D < 1$ derived from Daligault²⁵ [their Eq. (2)],

$$\ell_{\text{mfp,ion}} \approx 24.655 \frac{N_D \lambda_D}{\ln(1 + 6.477 N_D)}. \quad (4)$$

Hence, omitting the numerical constants for clarity, the collision rate normalized by the plasma frequency scales as

$$\frac{\nu_{\text{coll}}}{\omega_p} \sim \frac{\ln(1 + N_D)}{N_D}.$$

For $N_D \ll 1$, one finds $\ln(1 + N_D) \approx N_D$ that implies a ratio ~ 1 , indicative of a strongly collisional regime. Conversely, at large N_D , the factor $\ln(1 + N_D)/N_D \approx N_D^{-1}$, indicating that collisions are negligible compared to collective oscillations, thus identifying a collisionless regime. This direct dependence on N_D demonstrates why the plasma parameter is an excellent metric for gauging collisionality. In our simulations, we find that the ion plasma parameter N_D develops only a

modest precursor increase ahead of the shock, while the largest enhancement remains confined to the narrow transition layer, so throughout the analysis, we use the far-upstream, unperturbed value of N_D to characterize the collisionality.

C. Shock width dependence on the plasma parameter

To vary the ion plasma parameter, we scan the density n_0 together with the ion temperature T_i . Neither n_0 nor T_i showed a systematic correlation with the shock width, holding N_D constant; their influence was only observed when N_D varied.²⁶ Following Sorasio *et al.*²² and Fiuza *et al.*,²⁷ we initialize hot electrons and cold ions, a choice that preserves an electron-dominated sound speed $c_s \simeq \sqrt{k_B T_e/m_i}$ and therefore speeds up shock formation. In the collisional regime ($N_D \lesssim 1$), we keep $T_e/T_i \approx 400$, which leaves c_s governed by electrons while retaining a finite T_i that produces measurable ion–ion scattering. Achieving the collisionless limit demands $N_D \gg 1$; since $N_D \propto T_i^{3/2}$, we raise T_i (and hence lower the ratio) to $T_e/T_i \approx 9$. This still ensures electron control of c_s , yet boosts T_i enough to reach $N_D \gtrsim 10^2$, yielding a genuinely collisionless shock.

In Fig. 3, we plot the measured shock widths, normalized by the upstream mean-free-path l_{mfp} and the injection Mach number M_{inj} , as a function of the upstream ion plasma parameter N_D . Each data point corresponds to a distinct OSIRIS run, as is given in Table I. At low N_D (strong collisional coupling), the shock widths are nearly independent of the value of N_D , and are of the order of a few l_{mfp} . This is the expected result in the collisional regime.⁹ At higher values of N_D , $N_D > 1$, the shock width drops below l_{mfp} , and is $\propto N_D^{-1}$, as expected theoretically.¹² This result is consistent with the expectation in the collisionless regime. In the intermediate ($N_D \approx 1$) regime, we find a smooth, gradual transition between these two asymptotic regimes.

In Fig. 3, we also plot the asymptotic scalings derived by Bret and Pe'er.⁹ The horizontal dashed line $\ell/(M_{\text{inj}} \cdot l_{\text{mfp}}) = 1$ and the slanted dashed curve $\ell/(M_{\text{inj}} \cdot l_{\text{mfp}}) = A \ln N_D/N_D$, where the dimensionless prefactor A was not fixed in the original theory. A least-squares fit to our measurements yields $A \simeq 0.7$; the red dashed line in the figure adopts this value. The measured points track the two analytic branches in their respective limits and exhibit a smooth bridge around $N_D \sim 1$, thereby validating the crossover scenario proposed by Bret and Pe'er.⁹ We note, however, that this value of A is mainly constrained by the more collisionless runs with $R_T = 9$, and it may still depend on additional parameters such as the temperature ratio R_T . In particular, the narrower N_D range available for the $R_T = 400$ sequence, together with the expected breakdown of the asymptotic scalings close to $N_D \sim 1$, does not allow us to robustly determine a separate prefactor for the $R_T = 400$ cases.

IV. ANALYSIS OF RESULTS AND COMPARISON WITH THEORY

Our findings are aligned with previous theoretical works and expand them to the intermediate region, where analytical approximations are invalid. In particular, Ref. 9 proposed that the ion plasma parameter N_D can act as a toggle for shock dissipation mechanisms: for $N_D < 1$, collisions dominate and the Mott–Smith formalism¹⁰ with the BGK collision term is appropriate; for $N_D > 1$, one transitions smoothly to a collisionless shock picture reminiscent of Tidman's analysis.¹²

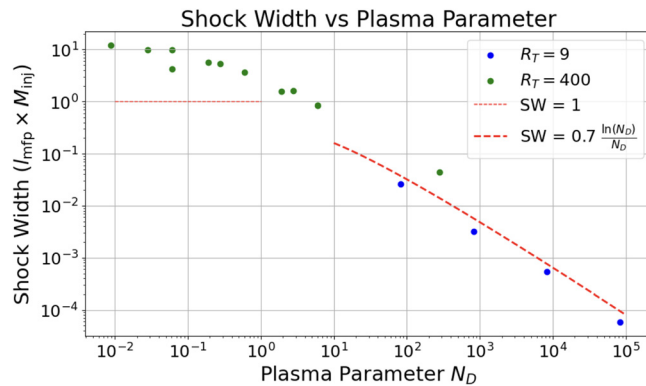


FIG. 3. Measured shock width (in units of the upstream mean-free-path times, the injection Mach number M_{inj}) vs the ion plasma parameter N_D . Data points are from multiple runs spanning densities 10^{16} – 10^{22} cm^{-3} and various temperature ratios. Around $N_D \sim 1$, shock widths begin decreasing below the collisional scale, indicating a shift to collisionless mediation. The horizontal dashed line $\ell/(M_{\text{inj}} \cdot l_{\text{mfp}}) = 1$ and the slanted dashed curve $\ell/(M_{\text{inj}} \cdot l_{\text{mfp}}) = 0.7 \ln N_D/N_D$ are the collisional and collisionless scalings, respectively, predicted by Bret and Pe'er.⁹

The Mott–Smith approach treats the ion distribution function as a superposition of upstream and downstream drifting Maxwellians, with the shock thickness scaling as a few mean-free-paths.¹⁰ Our simulation data at low plasma parameter N_D follow this pattern, with ℓ/l_{mfp} in the range of a few mean-free-paths. This is consistent with the fact that the shock formation is driven mainly by binary collisions.

Tidman¹² analyzed high-Mach-number electrostatic shocks and showed that the width scales with the ion Debye length (multiplied by factors dependent on the Mach number and logarithmic corrections). In the collisionless limit, l_{mfp} becomes irrelevant, so $\ell \ll l_{\text{mfp}}$. Our high- N_D runs exhibit precisely such behavior; we detect narrower shock fronts, often accompanied by electrostatic potential barriers that reflect a noticeable fraction of incoming ions.

Thus, the bridging scenario proposed by Bret and Pe'er⁹ is borne out in our simulations: the plasma parameter N_D is indeed a reliable metric for indicating when to expect collisionless or collisional behavior, at least in this unmagnetized, nonrelativistic context.

V. CONCLUSION

We have presented a series of 2D PIC simulations with the OSIRIS code, in which two interpenetrating electron–ion plasmas form shocks across a broad spectrum of collisionalities. By systematically varying density, temperature, and relative flow velocity, we probed the transition from collisional to collisionless shock formation. Our results corroborate recent theoretical predictions that this transition is governed by the ion plasma parameter, N_D . When $N_D < 1$, the shock width is controlled by classical collisional processes, matching Mott–Smith-like theories. Above $N_D \approx 1$, collisionless collective effects dominate, yielding narrower shocks consistent with Tidman’s work on electrostatic shocks.

Several caveats accompany our study. First, although OSIRIS solves the full set of electromagnetic Maxwell equations, we operate in a quasi-1D, low-Mach-number regime where electrostatic modes are expected to dominate; under these conditions, we can safely neglect the small electromagnetic corrections. The simulations also remain two-dimensional and unmagnetized, so external fields and fully three-dimensional electromagnetic instabilities—which may modify the transition—are deliberately excluded. Second, we adopt a fixed electron-to-ion temperature ratio ($T_e/T_i \simeq 400$ for collisional runs and $\simeq 9$ in the most collisionless cases) to keep the sound speed electron-dominated; alternate choices could shift the precise value of N_D at which the crossover occurs. Future work should therefore relax these same simplifications: extend the calculations to fully three-dimensional, magnetized geometries so as to capture electromagnetic instabilities, scan a wider range of temperature ratios (and hence of the sound speed), and include additional physics such as multiple ion species or relativistic drifts—each of which will test whether the N_D scaling inferred here remains robust once the neglected effects are reinstated. Another important extension will be to test whether the same N_D scaling holds for shocks generated by pressure imbalances between left and right states, rather than by symmetric counterstreaming slabs. These extensions would also complement and extend into the non-relativistic limit the relativistic unmagnetized-shock studies in which the Weibel instability governs the transition.¹⁷

The collisional–collisionless crossover studied here naturally occurs when a dense shock leaves the stellar interior and enters a much lower-density medium.²⁸ Numerical models of trans-relativistic supernova shock breakouts, for instance, show that the front is

collisional while still inside the envelope but turns collisionless immediately after breakout,²⁹ once the upstream density drops and the ion plasma parameter N_D surges. A similar evolution is inferred for coronal mass-ejection (CME) shocks in the solar corona: radio and white-light diagnostics indicate a collisional front at $R \lesssim 6 R_\odot$, which is replaced by a collisionless discontinuity beyond $R \gtrsim 10 R_\odot$ as the proton mean-free-path exceeds the shock width.³⁰ In that newly collisionless phase, the self-generated electromagnetic fields can accelerate a fraction of upstream ions to non-thermal energies, producing high-energy radiation and neutrinos.²⁹ By demonstrating that the shock width obeys a universal N_D scaling across the transition, our simulations provide a practical diagnostic for pinpointing where such particle acceleration should be triggered in both astrophysical outflows and laboratory experiments that emulate breakout conditions.

ACKNOWLEDGMENTS

We gratefully acknowledge helpful discussions with Lorenzo Sironi, whose insights helped us finish this project. Antoine Bret also acknowledges support from the Ministerio de Economía y Competitividad of Spain (Grant Nos. PID2021-127322OB-I00 and PID2024-157933OA-I00). Kevin M. Schoeffler is supported by the German Science Foundation (DFG) within the Collaborative Research Center SFB1491.

AUTHOR DECLARATIONS

Conflict of Interest

The authors have no conflicts to disclose.

Author Contributions

Yossef Nissim Kindi: Data curation (lead); Formal analysis (lead); Investigation (lead); Writing – original draft (lead). **Asaf Pe'er:** Conceptualization (lead); Investigation (equal); Project administration (lead); Supervision (lead); Writing – review & editing (lead). **Antoine Bret:** Conceptualization (lead); Investigation (equal); Supervision (supporting); Validation (equal); Writing – review & editing (equal). **Luís O. Silva:** Resources (equal); Software (equal); Supervision (supporting); Validation (supporting); Writing – review & editing (supporting). **Kevin M. Schoeffler:** Resources (equal); Software (equal); Supervision (equal); Writing – review & editing (supporting).

DATA AVAILABILITY

The data that support the findings of this study are available from the corresponding author upon reasonable request.

REFERENCES

- ¹Y. B. Zel'dovich and Y. P. Raizer, *Physics of Shock Waves and High-Temperature Hydrodynamic Phenomena*, Dover ed. (Dover Publications, Mineola, NY, 2002).
- ²L. D. Landau and E. M. Lifshitz, “Course of theoretical physics,” in *Fluid Mechanics*, 2nd ed. (Pergamon Press, Oxford, 1987), Vol. 6.
- ³O. Buneman, *Phys. Fluids* **7**, S3 (1964).
- ⁴R. Z. Sagdeev, *Rev. Plasma Phys.* **4**, 23 (1966), Originally published in Russian in 1964; available at <https://ui.adsabs.harvard.edu/abs/1966RvPP....4...23S/exportcitation>.
- ⁵S. D. Bale, F. S. Mozer, and T. S. Horbury, *Phys. Rev. Lett.* **91**, 265004 (2003).
- ⁶S. J. Schwartz, E. Henley, J. Mitchell, and V. Krasnoselskikh, *Phys. Rev. Lett.* **107**, 215002 (2011).

- ⁷A. Bamba, R. Yamazaki, M. Ueno, and K. Koyama, *Astrophys. J.* **589**, 827 (2003).
- ⁸A. Bret, A. Stockem, F. Fiuza, C. Ruyer, L. Gremillet, R. Narayan, and L. O. Silva, *Phys. Plasmas* **20**, 042102 (2013).
- ⁹A. Bret and A. Pe'er, *J. Plasma Phys.* **87**, 905870204 (2021).
- ¹⁰H. M. Mott-Smith, *Phys. Rev.* **82**, 885 (1951).
- ¹¹P. L. Bhatnagar, E. P. Gross, and M. Krook, *Phys. Rev.* **94**, 511 (1954).
- ¹²D. A. Tidman, *Plasma Physics and the Problem of Controlled Thermonuclear Reactions* (Addison-Wesley, 1967), pp. 58–81.
- ¹³We model a hydrogen (proton–electron) plasma, so the ion charge state is $Z = 1$.
- ¹⁴J. Zheng, C. X. Yu, and Z. J. Zheng, *Phys. Plasmas* **6**, 435 (1999).
- ¹⁵R. A. Fonseca, L. O. Silva, F. S. Tsung, V. K. Decyk, W. Lu, C. Ren, W. B. Mori, S. Deng, S. Lee, T. Katsouleas, and J. C. Adam, in *Computational Science—ICCS 2002*, Lecture Notes in Computer Science Vol. 2331, edited by P. M. A. Sloot, A. G. Hoekstra, C. J. K. Tan, and J. J. Dongarra (Springer, Berlin, Heidelberg, 2002), pp. 342–351.
- ¹⁶R. Fonseca, F. Tsung, S. Deng, and C. Ren, in *APS Division of Plasma Physics Meeting Abstracts* (APS, 2005), Vol. 47, p. CP1.066.
- ¹⁷M. Fiore, F. Fiuza, M. Marti, R. A. Fonseca, and L. O. Silva, *J. Plasma Phys.* **76**, 813 (2010).
- ¹⁸T. Takizuka and H. Abe, *J. Comput. Phys.* **25**, 205 (1977).
- ¹⁹K. Nanbu, *Phys. Rev. E* **55**, 4642 (1997).
- ²⁰F. Peano, M. Marti, L. O. Silva, and G. Coppa, *Phys. Rev. E* **79**, 025701 (2009).
- ²¹F. Pérez, L. Gremillet, A. Decoster, M. Drouin, and E. Lefebvre, *Phys. Plasmas* **19**, 083104 (2012).
- ²²G. Sorasio, M. Marti, R. Fonseca, and L. O. Silva, *Phys. Rev. Lett.* **96**, 045005 (2006).
- ²³F. F. Chen, *Introduction to Plasma Physics and Controlled Fusion* (Springer, 2016).
- ²⁴On a subset of runs, we also repeated the simulations with collisions restricted to ion–ion scattering only and found that the resulting shock widths agree with those obtained using the full multi-species collision operator.
- ²⁵J. Daligault, *Phys. Rev. Lett.* **108**, 225004 (2012).
- ²⁶We numerically checked that a four orders of magnitude change in the density between two points with similar plasma parameters results in a similar shock width. Similarly, a two orders of magnitude variation in the temperature does not affect the shock width for similar plasma parameters.
- ²⁷F. Fiuza, A. Stockem, E. Boella, R. A. Fonseca, L. O. Silva, D. Haberberger, S. Tochitsky, W. B. Mori, and C. Joshi, *Phys. Plasmas* **20**, 056304 (2013).
- ²⁸A. Bret and A. Pe'er, *J. Plasma Phys.* **84**, 905840311 (2018).
- ²⁹K. Kashiwama, K. Murase, S. Horiuchi, S. Gao, and P. Mészáros, *Astrophys. J. Lett.* **769**, L6 (2013).
- ³⁰V. Eiselevich and M. Eiselevich, *Astrophys. J.* **761**, 68 (2012).